Fraud Attack & Defence Simulator with AI

An adversarial multi-agent red-team/blue-team framework with 3-tier RAG & adaptive learning

Motivation & Research Gap

- Three autonomous agents battle iteratively - **Red Team** attacks, **Blue Team** defends, **Judge** scores - with a 3-tier RAG architecture driving adaptation each round.
- Domain-agnostic design - demonstrated with web security, extensible to other fraud detection for **Sun Life Financial**.
- Existing tools like MITRE CALDERA and CyberBattleSim use static rule sets with **no persistent cross-engagement memory**.
- No open-source framework models the **co-evolution** of attacker and defender strategies using local LLMs on commodity hardware.
- Agents run entirely on **local hardware** via Ollama - no cloud dependency, no data leaves the machine.

Methodology

- **Agile** - monthly sprints covering RAG setup, agent prompt engineering, and MCP integration, with regular supervisor and industry mentor reviews.
- **Adversarial cycle** - each round is a controlled experiment tracking attack novelty, defence effectiveness, and strategy diversity across agents. Judge scores are persisted to ChromaDB after every round.
- **Theorise-before-build** - all major architectural decisions were validated before implementation.

Key Technologies

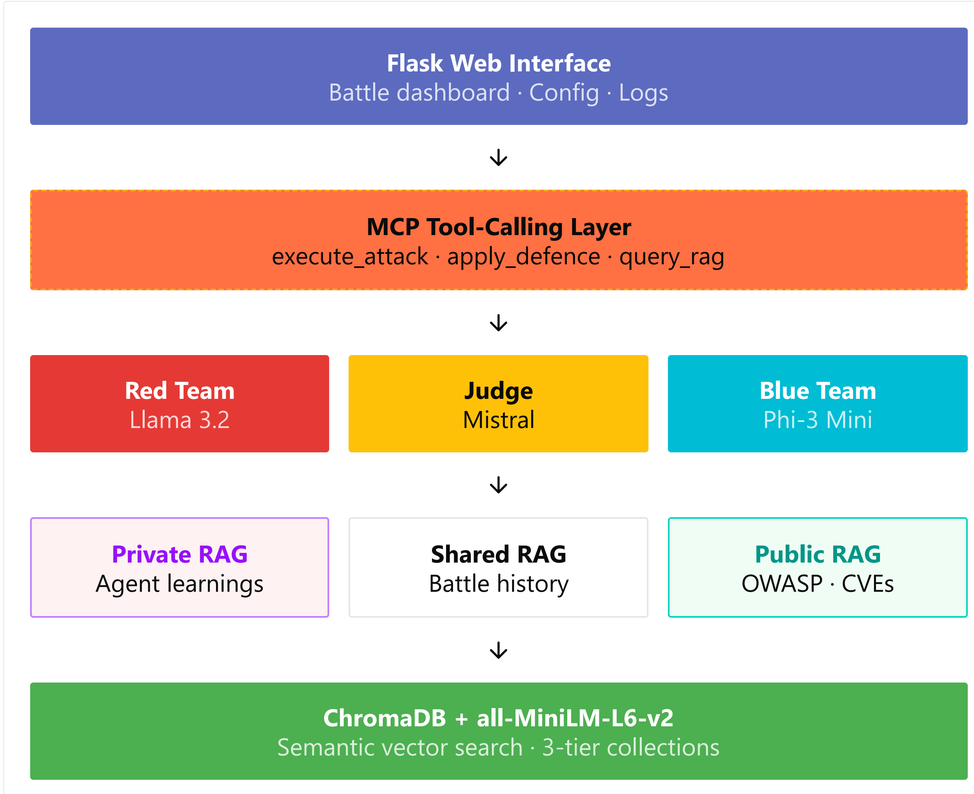
Python · Flask · AWS EC2 · ChromaDB · Ollama · MCP · SentenceTransformer · Linux

Vector Embeddings

Text Chunk → MiniLM-L6-v2 → [0.23, -0.14...] → ChromaDB

- Locally cached - no API calls
- Chosen over Ollama embeddings for efficiency and display
- Cosine similarity search returns top-k chunks as agent context per round
- 384-dimensional vectors stored across three scoped ChromaDB collections - private, shared, and public knowledge.

System Architecture



Battle Loop

Red Team Attack → **Blue Team Defence** → **Judge Verdict** → **Metacognition** → **Memory Update** → **Next Round**

- **Red Team Attack**: Queries private RAG, generates attack with mandatory diversity rotation
- **Blue Team Defence**: Queries private RAG, constructs targeted mitigation response
- **Judge Verdict**: Scores novelty and effectiveness, penalises repeated attack types
- **Metacognition**: Each agent self-evaluates its performance before memory is written
- **Memory Update**: Outcomes written to shared RAG, learnings stored in private collections
- **Next Round**: Agents adapt using enriched context, loop repeats

↺ repeats each round

Agent Roles

Red Team Agent
Generates diverse attack strategies with mandatory type rotation. Queries private RAG for novel patterns and adapts tactics based on Judge novelty scoring to maximise strategic coverage.

Blue Team Agent
Constructs defences using RAG-retrieved context from shared battle history and its own private collection, building a compounding library of successful mitigations each round.

Judge Agent
Scores rounds on novelty, effectiveness, and escalation. Writes outcomes to shared ChromaDB, penalises repeated attack types, and triggers metacognition prompts before memory is written.

System Configuration

- **Model selection** - swap any agent's LLM at runtime to change reasoning style and creativity
- **Temperature** - higher increases attack novelty; lower produces more consistent defences
- **RAG top-k** - controls ChromaDB results per query; higher gives broader context at the cost of inference time
- **Attack domain** - switching the knowledge base shifts threat context from web security to fraud detection with no code changes

3-Tier RAG Architecture

Tier 1 - Private Agent Collections
Per-agent strategic learnings - not shared. Forms each agent's evolving "brain" across rounds.

Tier 2 - Shared Battle History
Full round outcomes visible to all agents. Enables cross-agent pattern recognition over time.

Tier 3 - Public Knowledge Base
OWASP guidelines, CVEs, fraud patterns. Static reference corpus seeded at initialisation.

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